

Broker API Options for the Quantitative Intraday Project

Where to get live and historical NSE data for 15-minute prediction, signal generation and paper trading

Note: Pricing and policy details below reflect the information supplied for this project and change often. Confirm current plans, limits and requirements on each provider's official developer page before you sign up or build around them.

1. Comparison at a glance

Provider	Live market data	Historical data	API cost	Fit for this project
Angel One SmartAPI	Yes	Yes	Free API	Very suitable
Upstox API	Yes (WebSocket)	Yes	Free access to trading and market-data APIs	Very suitable
DhanHQ	Yes	Yes	Trading API Rs. 0; Data API currently Rs. 499	Very suitable
Zerodha Kite Connect	Yes	Yes	Paid API plan	Suitable

2. Provider details

2.1 Angel One SmartAPI (free)

- Advertised as free of cost, with real-time market data and trading integration, and Python support.
- Attractive if your main goal is: live data, then 15-minute candles, then ML prediction, then dashboard.
- **Important:** since 1 April 2026, Angel One requires API *order execution* to originate from a registered static IP. This matters only if you later place real orders; it does not affect a paper-trading project.

2.2 Upstox API (free access)

- Currently states Rs. 0 free access to trading and market-data APIs, with real-time data over WebSocket.
- Covers what the project needs: live stock price, OHLC, volume, NIFTY 50, India VIX and intraday data.
- Offers a free read-only **Analytics Token**, valid for one year, for market-data and historical-data access. This is convenient for a research project because it cannot place orders.

2.3 DhanHQ (free trading API, paid data API)

- Trading APIs: Rs. 0. Data APIs: currently Rs. 499.
- Data APIs include real-time market feeds and historical OHLC, volume and open-interest data.
- Has an official Python library, which makes integration with the project straightforward.

2.4 Zerodha Kite Connect (paid)

- Provides live and historical data, but requires a paid API plan.

- Well documented and widely used; a reasonable choice later if you want a paid, stable feed.

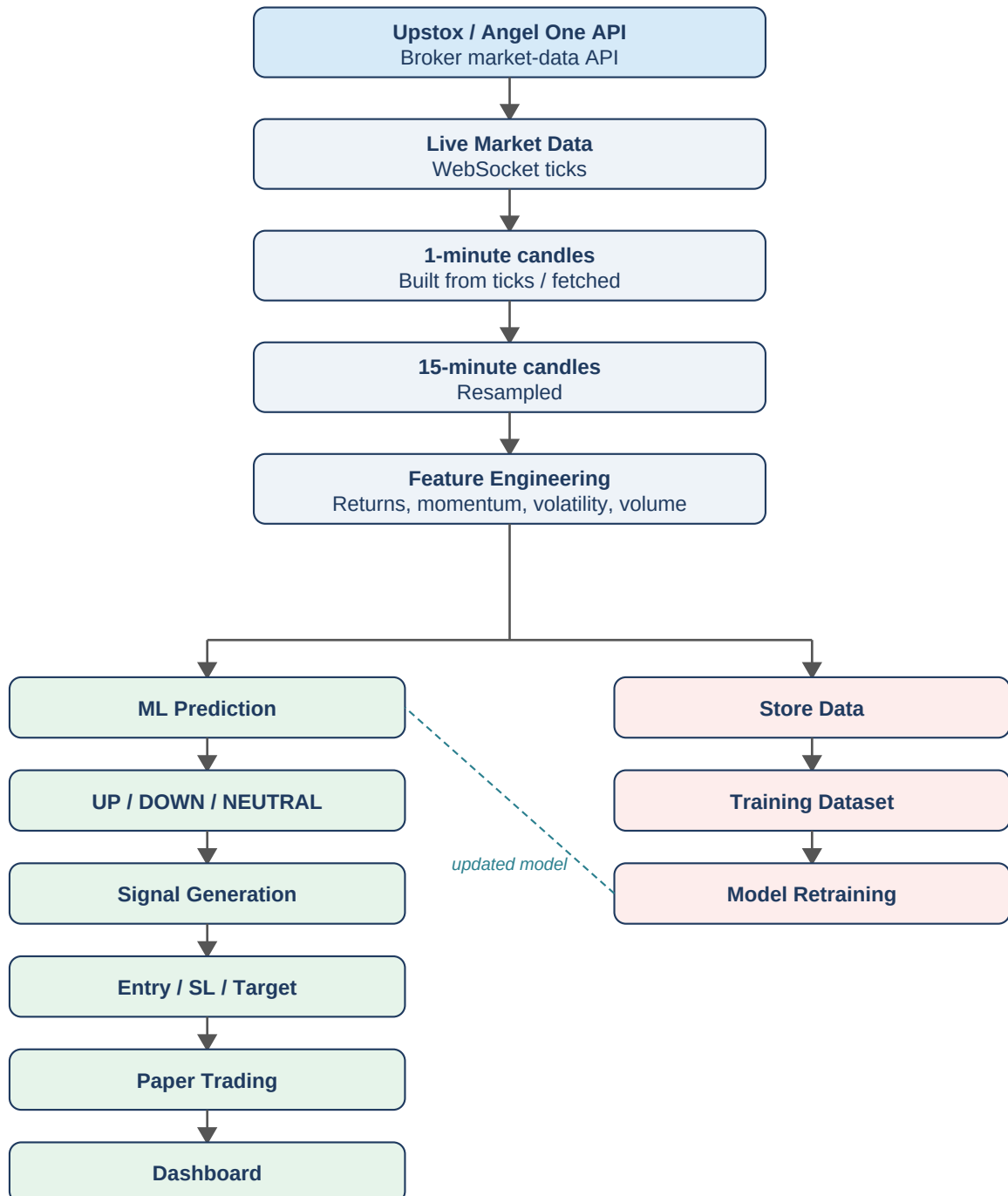
3. Recommendation

First choice: Angel One SmartAPI or Upstox. You do not need to pay for API access initially. Start with one, build the pipeline, and add a second provider only if you need a cross-check or longer history.

Choose	When
Upstox	You want a free WebSocket feed and a read-only token for historical data with no order-placement risk.
Angel One SmartAPI	You want a free API with Python support and may later connect order execution.
DhanHQ	You are fine paying Rs. 499 for a dedicated Data API and want its official Python library.
Zerodha Kite Connect	You already use Zerodha and accept a paid plan.

4. Suggested project setup

The flow below shows how a broker API feeds the whole pipeline, from live data through paper trading. Live bars go both to the prediction engine and into storage, so the training dataset grows and the model can be retrained over time.



5. How the pieces map to the project document

Project requirement	What to pull from the API
Stock OHLCV (e.g. RELIANCE)	1-minute candles, resampled to 5 and 15 minutes
Market context	NIFTY 50, BANK NIFTY and a sector index
Volatility regime	India VIX
Backtesting dataset	As much historical intraday data as the provider allows
Live dashboard	WebSocket ticks or polling, aggregated into 15-minute bars

6. Practical tips

- **Build your own history.** Broker APIs often limit how far back intraday candles go, so store every bar you receive from day one. This becomes your training dataset.
- **Avoid lookahead bias.** Use only completed 15-minute bars when generating a signal, and align index and VIX data to the same timestamps.
- **Paper trade first.** Apply the stop-loss, target, position sizing, costs and slippage from the concept note to simulated orders before any real money is involved.
- **Respect rate limits and market hours.** NSE trades 9:15 to 15:30 IST; handle reconnects, missing bars and holidays.
- **Protect credentials.** Keep API keys, secrets and tokens in environment variables, never in code or a public repository.

7. Sign-up checklist

- Open an account with your chosen broker (Angel One or Upstox for the free route).
- Create an app in the developer portal and note the API key and secret.
- For Upstox, generate the read-only Analytics Token if you only need data.
- Install the official Python SDK and test one historical-candle call for RELIANCE.
- Confirm the current historical-data limits (intraday range, candle intervals, request rate) in the official docs.
- Set up storage (CSV, SQLite or Parquet) and start logging bars daily.